

Linux Permission And Security Theory TASK – 3

- 1. Create two new users on your Linux system named 'musicfan' and 'moviebuff', then create a new group called 'playlist' and add both users to this group.**

Ans :

- # Create the users
- sudo useradd -m musicfan
- sudo useradd -m moviebuff

- # Create the group
- sudo groupadd playlist

- # Add both users to the group
- sudo usermod -aG playlist musicfan
- sudo usermod -aG playlist moviebuff

- 2. Inside your home directory, create a folder called 'spotify_shared'. Set its permissions so only users in the 'playlist' group can read and write files inside it, but others cannot access it.

Hint: Use chown and chmod with the group ownership and permission bits.**

Ans :

- # Create the directory
- mkdir ~/spotify_shared

- # Change the group ownership to playlist
- sudo chown :playlist ~/spotify_shared

- # Set permissions:
- # Owner: rwx
- # Group: rwx
- # Others: ---
- chmod 770 ~/spotify_shared

3. Place a text file named 'top_songs.txt' inside 'spotify_shared'. Change its ownership so that 'musicfan' is the owner and 'playlist' is the group, then set permissions so only the owner can edit, group members can read, and others have no access.

Ans :

- # Create the file
- touch ~/spotify_shared/top_songs.txt
- # Set owner to musicfan and group to playlist
- sudo chown musicfan:playlist ~/spotify_shared/top_songs.txt
- # Set permissions:
- # Owner: read/write
- # Group: read only
- # Others: no access
- chmod 640 ~/spotify_shared/top_songs.txt

4. Simulate a scenario where 'moviebuff' tries to access 'top_songs.txt'. Switch to the 'moviebuff' user and attempt to open and edit the file. Describe what happens and explain why based on the permissions you set.

Ans :

- # Switch to moviebuff
- su – moviebuff

- # Read the file
- cat /home/<your_user>/spotify_shared/top_songs.txt

- # Try to append text to the file
- echo "New Song" >>
/home/<your_user>/spotify_shared/top_songs.txt

The cat command should succeed (assuming moviebuff can traverse the directory and the path is correct), because:

- moviebuff belongs to the playlist group.
- The file permissions are 640 (rw-r-----).
- Group members have read (r) permission.

Why it fails:

- File owner: musicfan
- Group: playlist
- Permissions: 640 (rw-r-----)